Does it think about the relevant nodes and edges at the relevant times?

Partly, and the part that is new is the EDGES. Only the leak-free model isolates the known neighbours that force the answer: 16 justifies_next features against 2, and best F1 0.486 vs 0.162 at layer 8. The frontier is a wash (~0.4 in both).

Two checkpoints identical in every hyperparameter but --shuffle-adj, each analysed on its own distribution. 53,200 positions from 3,000 N=24 instances; d_sae=4096; dictionaries trained to a measured sparsity target rather than a guessed one.

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING BEGIN_ROUND SOLVED UNRESOLVED	*section + round markers*
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3	*the q colours*
10-33	N0 N1 ... N23	*one id per node, no structure in the id*

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

The graph section is a neighbour table, not an image: node, then its degree neighbours. THE ORDER WITHIN A ROW IS THE LEAK of finding 16.

THIS INSTANCE, 204 TOKENS (seed 50000000038)

HINTS	BEGIN_HINTS N0 COLOR_2 N1 COLOR_1 N4 COLOR_1 N5 COLOR_1 N7 COLOR_2 N9 COLOR_0 ... (23 tokens)
GRAPH	BEGIN_GRAPH N0 N17 N21 N6 N13 N9 N1 N19 N12 N6 N7 N14 ... (145 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N3 COLOR_3 N21 COLOR_3 N23 COLOR_3 BEGIN_ROUND N6 COLOR_0 N15 COLOR_1 ... (36 tokens)

Loss is on the REASONING section only: 8 rounds, 13 deductions, 26 target tokens. Everything before BEGIN_REASONING is the problem statement.

One real reasoning trace

11 hints of 24 nodes, then 8 rounds of simultaneous deduction. Hints: N0=2 N1=1 N4=1 N5=1 N7=2 N9=0 N10=2 N12=0 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N3 COLOR_3 N21 COLOR_3 N23 COLOR_3
  3 deduced
BEGIN_ROUND          N6 COLOR_0 N15 COLOR_1
  2 deduced
BEGIN_ROUND          N11 COLOR_3 N13 COLOR_0
  2 deduced
BEGIN_ROUND          N18 COLOR_2 N20 COLOR_0
  2 deduced
... 4 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N3, neighbours [17, 4, 10, 19, 5]

colours known before this round:

```
   [0, 1, 2]  (3 = q-1)
```

only colour left: 3

```
so the trace emits  N3 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

T=8 rounds, the 90th percentile of 40 draws from this config (range 1-10); seed 50000000038, shuffle_adj=True; evidence from sudoku_tokens.evidence_rounds (pre-round state, as DD saw it)

First, why the previous run was void

49702811 exited 0:0 and looked healthy (FVE 0.97-0.99). Its l1 search never ran: it only ever RAISED l1, and accepted as soon as L0 fell BELOW the target, so every fit stopped on iteration 1 at the start value.

layer	L0 void	L0 now	justif void	now	nbr void	now
0	2.0	5.9	0.000	0.000	0.000	0.000
2	15.6	64.4	0.091	0.167	0.000	0.000
4	13.6	55.3	0.095	0.118	0.000	0.000
6	15.8	60.8	0.201	0.166	0.392	0.574
8	26.7	56.9	0.239	0.486	0.197	0.651
10	28.6	61.4	0.232	0.322	0.174	0.564
12	32.3	71.6	0.257	0.358	0.233	0.352

WHAT IT COST

Ground truth: 55.8 DD predicates are true at once per position.

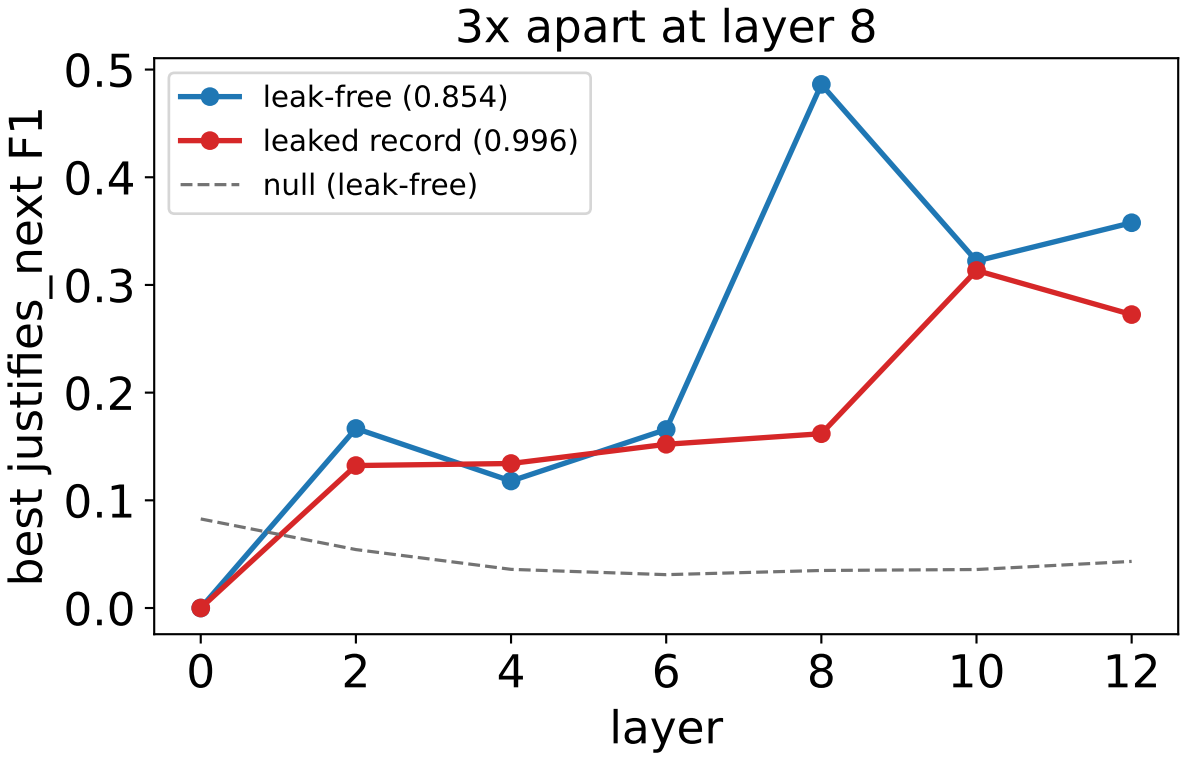
The void dictionaries sat at L0=2-16, with ~130 of 4096 features alive at layer 0. A dictionary that sparse cannot carry the code, so its zeros on the edge families described the DICTIONARY, not the model.

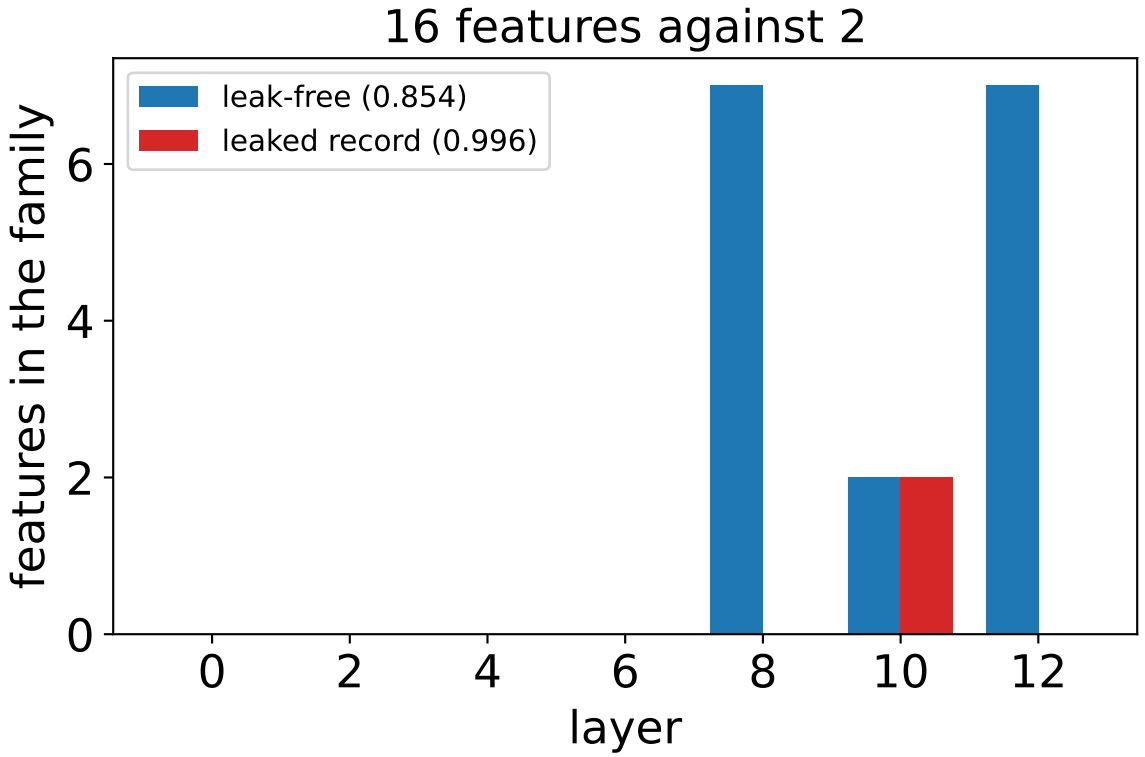
Every DD signal was understated. None of (a)-(d) was tested.

Fixed in 83714d7: the search now brackets the target from both sides, bisects in log space, returns the fit CLOSEST to the target, and flags a miss either way. Layer 0 is the one layer that still misses -- it is the token+position embedding, whose true sparsity is ~2 at FVE=1.000 -- and it now says so on the page instead of claiming a hit.

The edges that force the answer

justifies_next[u] = u is a KNOWN neighbour of the node being emitted, i.e. one of the edges doing the forcing.



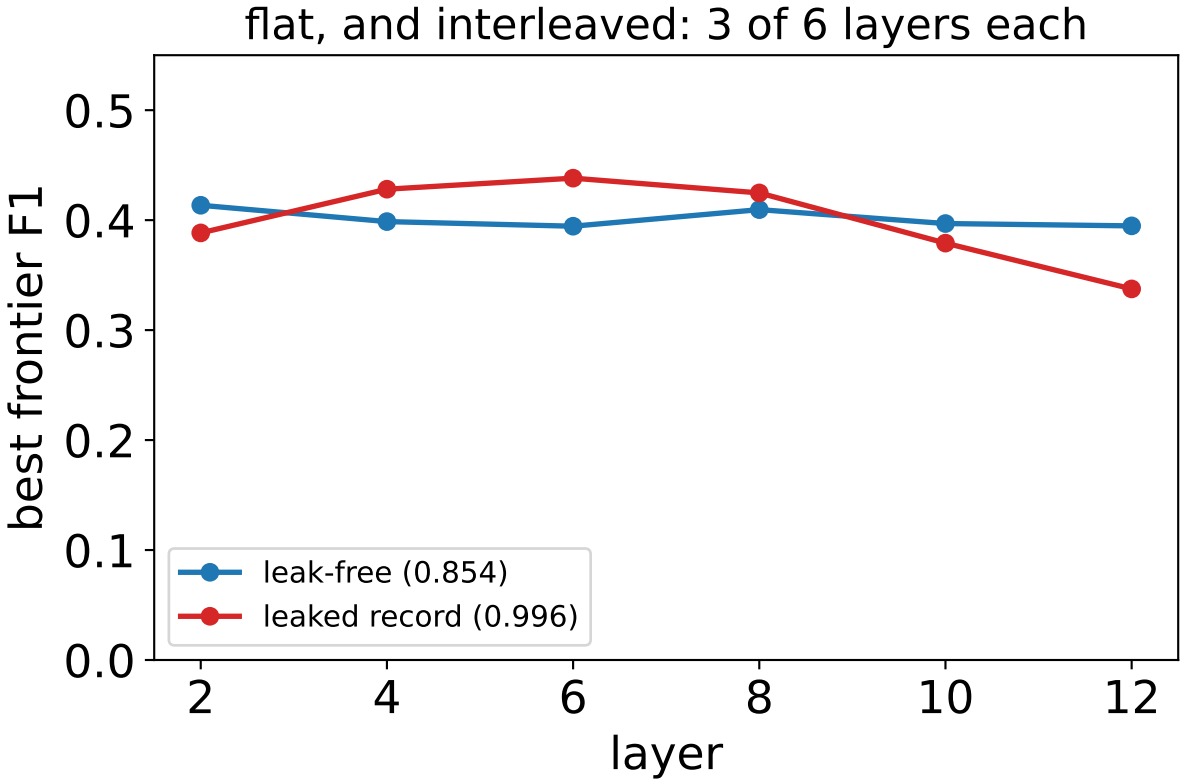


Both models find the node to emit (next_node, every layer) and its neighbourhood (nbr_of_next, layers 8+).
What separates them is the KNOWN subset of that neighbourhood -- the part that actually determines the colour.
The leaked model could read the forced colour off the sorted order without ever representing which
neighbours were known, and largely did not represent it.

target L0=64; F1 at the 0.3 cut and above a 19-roll position null; the robustness pass at L0=32 keeps the layer-8 and layer-12 ordering

What does NOT separate them

The frontier is the DD predicate itself, and I predicted it would be higher for the leak-free model at every layer.



layer	leak-free	leaked	higher
2	0.414	0.388	leak-free
4	0.399	0.428	leaked
6	0.394	0.438	leaked
8	0.410	0.425	leaked
10	0.397	0.379	leak-free
12	0.395	0.337	leak-free

Both sit at 0.39-0.44 at every layer. Whatever the frontier features are tracking, the shortcut did not change it -- so 'computes the frontier' does not distinguish a model that reasons from one that decodes token order. The bookkeeping families I expected to dominate (known, color) are nearly absent in both: 0-19 features of ~4000.

target L0=64, layers above 0; nulls differ slightly between arms (0.031-0.077), so the above-null margin mildly favours the leak-free model where the raw F1 does not

Pre-registrations, as resolved

Written in the sbatch header before the job ran. Two of four held.

(a)	No layer-0 next_node features in the leak-free model; the leaked one keeps them	**FALSIFIED**	23 vs 24 features -- identical. Layer 0 is the embedding, and at pre_color positions the current token IS next_node, so 'current token = N7' matches the predicate for free, in both models.
(b)	The leak-free model has justifies_next features mid-stack; the leaked one fewer or none	**CONFIRMED**	16 features vs 2, and best F1 0.486 vs 0.162 at layer 8. The headline result.
(c)	best_frontier_f1 higher for the leak-free model at every layer above 0	**FALSIFIED**	Higher at 3 of 6 layers; both flat at 0.39-0.44. The frontier does not separate them.
(d)	Neither dictionary is DD-shaped: known/color bookkeeping outnumbers frontier+justifies	**FALSIFIED**	Backwards. Bookkeeping is nearly absent (0-19); next_node dominates everything (55-218 per layer). The SAE does not mirror the state distribution.

My record on mechanism predictions in this project is now 4 of 9; (a), (c) and (d) were all wrong in the same way -- I expected the leak to be the only mechanism

What this does and does not show

IT DOES SHOW

The leak-free model represents the forcing edges and the leaked one mostly does not, at a sparsity set by the ground truth's own complexity rather than guessed.

The ordering survives the robustness pass at L0=32 at layers 8 and 12.

IT DOES NOT SHOW

That the model runs DD. next_node dominates every dictionary, and a 0.486 F1 on one family at one layer is structure, not an algorithm.

One seed, one checkpoint per arm, one instance distribution. best_f1 is a max over 256 predicates; the family counts depend on the F1>=0.3 cut.

OWED NEXT. Split the layer-0 next_node features by position kind to confirm the token-identity explanation for (a) (the predicate set already carries kind_is_node). Then the same analysis on a second seed, since every number here is n=1.